

Digital Farming Assistant

Purpose

The goal of the Digital Farming Assistant is to empower farmers through a digital platform that uses data-driven technologies to enhance agricultural productivity and decision-making. The reason why we think the problem is important is because of general limitations found in farming apps. Such as:

- **Digital Literacy Barriers:** Many marginal farmers struggle with app navigation, reading complex text prompts, or completing multi-step digital forms.
- **Lack of Hyper-Local Precision:** Soil profiles and micro-climates vary widely across small regions; generalized app data may not match specific field conditions.
- **Lack of connectivity:** Many of them do not have any kind of forum where people can share their concerns and get feedback on it.
- **Fragmented Ecosystems:** Separate apps handle crops, livestock, government sales, and weather, meaning users must switch between multiple platforms instead of using a single unified assistant.
- **Crop Disease Treatment & Prevention Advice**
- **Location-Based Weather Forecast**
- **Agricultural Weather Alerts**
- **Diagnosis History**

Objectives of this system includes:

- ❖ Providing AI/ML-based crop health detection and feedback.
- ❖ Offering real-time weather forecasts to aid crop planning.
- ❖ Maintaining a crop library with cultivation guidelines.
- ❖ Creating a community forum for farmers to share experiences and solutions.
- ❖ Simplifying user access with a minimal login system (name and phone number only).

Inception:

The Inception phase identifies the stakeholders, their needs, and their expectations from the proposed Digital Farming Assistant. The main stakeholders include farmers, agricultural experts, administrators, project sponsors, the development team, weather data providers, and government agriculture officers.

Farmers need easy access to crop diagnosis, farming information, weather updates, AI assistance, and expert support. Experts and government officers focus on accurate and verified agricultural information, while administrators focus on security, content management, and quality assurance. The development team and project sponsor focus on feasibility, project scope, quality, and successful completion.

This phase also identifies common and conflicting requirements among stakeholders so that they can be prioritized and combined into the final system requirements.

2.1 Identifying Stakeholders

ID	Name/Role	Organization	Position	Why They Are a Stakeholder	Expectations	Type of Interactions
S 1	Project Sponsor	East West University	Final academic authority, approves scope	Sponsors the project.	Successful implementation, academic quality and achievement of project objectives.	Reviews reports, approves scope.
S 2	Project Manager	Project Team	Leads execution and coordination	Lead the Development Team.	Timely completion, proper coordination and successful delivery.	Uses the system for management monitoring and coordination.
S 3	HR Consultant (Dr. Rajesh Chakraborty)	Shere Bangla Agricultural University	Advisory	Provides professional advice.	A practical and user-friendly system that addresses stakeholder needs.	Provides feedback, reviews requirements, and gives recommendations.
S 4	Development Team	EWU Students	Designs and develops system	Develops the entire system.	Clear requirements, technical feasibility, and successful system operation.	Directly develops, tests, updates, and maintains the system.
S 5	Farmer Representative	Farming Community	Represent end users	Represents the primary end users of the system.	Easy access to useful farming information and communication with other farmers.	Uses the app, provides feedback, and communicates through the farmer community.
S 6	Agricultural Expert	Agriculture Sector	Provides crop guidance knowledge	Provides reliable agricultural knowledge for the system	Accurate and useful agricultural information reaching farmers.	Provides/updates guidance and responds to agricultural queries through the platform.
S 7	Weather Data Provider	Meteorological Dept	Supplies forecast data	Provides weather information needed for	Accurate and timely presentation of weather information.	Provides weather data through data integration/API or regular updates.

				farming decisions.		
S 8	Government Agriculture Officer	Govt. of Bangladesh	Regulatory Oversight, Quality Assurance	Represents the government agricultural support.	A reliable system that supports farmers and complements agricultural services.	Reviews information, provides guidance, and may interact with farmers through the platform.

2.2 Identifying multiple viewpoints of the stakeholders

1. Project Sponsor

What does this stakeholder want from the system?

- A complete and functional **Digital Farming Assistant** that addresses a real agricultural problem.
- A system that satisfies the approved project objectives and requirements.
- A practical application that demonstrates meaningful software engineering concepts.
- A system that provides useful agricultural services to farmers.
- Successful integration of the major system features.

What features do they consider important?

- User registration and login.
- Bengali and English language support.
- Automatic location detection.
- AI-based crop health diagnosis.
- Crop disease treatment and prevention advice.
- Diagnosis history.
- Location-based weather forecasts.
- Agricultural weather alerts.
- Searchable crop information library.
- Offline crop information and diagnosis queue.
- Farmer community forum.
- Image sharing, comments, and likes.
- Notifications.
- AI chatbot for agricultural questions.

- Admin dashboard and user management.
- Content moderation.
- Security and activity logs.
- Regulatory oversight, quality assurance, and input verification.

What constraints or concerns do they have?

- The system must remain within the approved academic scope.
- The application should be fully functional and demonstrable.
- Agricultural information should be reliable.
- User data should be protected.
- AI-generated information should be presented responsibly.
- The project should be completed within the academic schedule.
- The system should meet the quality expectations of the university.

2. Project Manager

What does this stakeholder want from the system?

- A system with clearly defined and manageable requirements.
- Successful completion of all agreed functionality.
- Effective coordination between different system components.
- A maintainable and reliable application.
- Clear separation between farmer, expert, moderator, and administrator functions.

What features do they consider important?

- User registration and login.
- Role-based access.
- User management.
- AI crop diagnosis.
- Diagnosis history.
- AI agricultural chatbot.
- Weather and agricultural alerts.
- Crop information library.
- Offline functionality.
- Community forum.
- Notifications.
- Admin dashboard.

- Content moderation.
- Security and activity logs.
- Input verification and quality assurance.

What constraints or concerns do they have?

- Limited development time and resources.
- Requirements should remain clearly defined.
- External services such as weather and AI services may have availability limitations.
- Offline functionality must handle synchronization appropriately.
- Security requirements must be maintained throughout the system.
- The system should remain manageable as the number of users and records increases.

3. HR Consultant

Shere Bangla Agricultural University

What does this stakeholder want from the system?

- A practical agricultural assistance system that addresses real problems faced by farmers.
- Reliable crop health information.
- Appropriate treatment and prevention recommendations.
- An AI assistant that can provide useful agricultural information.
- A system that helps farmers understand agricultural problems in simple language.
- Proper verification of agricultural information before it becomes part of the system.

What features do they consider important?

- AI-based crop health diagnosis.
- Crop disease treatment and prevention advice.
- Searchable crop information library.
- AI agricultural chatbot.
- Diagnosis history.
- Farmer community forum.
- Agricultural expert involvement.
- Bengali language support.
- Input verification.
- Regulatory oversight and quality assurance.
- Weather and agricultural alerts.

What constraints or concerns do they have?

- AI-generated agricultural recommendations may not always be completely accurate.
- Treatment recommendations should be based on reliable agricultural knowledge.
- The system should clearly distinguish AI-generated guidance from verified expert information.
- Fertilizer recommendations should consider factors such as crop, field area, and available agricultural information.
- Harmful or incorrect agricultural advice should be prevented from being presented as verified information.
- Agricultural content should be reviewed or verified by appropriate knowledgeable stakeholders.

4. Development Team

What does this stakeholder want from the system?

- Clear and complete requirements.
- Well-defined functionality for each type of user.
- A system that can be developed, tested, maintained, and demonstrated successfully.
- Clearly defined interactions between farmers, experts, administrators, and other stakeholders.

What features do they consider important?

- Registration and login.
- Bengali/English interface.
- Location detection.
- AI crop diagnosis.
- Treatment and prevention advice.
- Diagnosis history.
- Weather forecasting.
- Weather alerts.
- Crop information library.
- Offline information and diagnosis queue.
- Community forum.
- Image sharing, comments, and likes.
- Notifications.
- AI chatbot.

- Admin dashboard.
- User management.
- Content moderation.
- Security and activity logs.
- Regulatory and input verification functions.

What constraints or concerns do they have?

- Limited project time and resources.
- AI diagnosis and chatbot responses may require external AI services.
- Weather information may depend on external data sources.
- Location detection requires user permission.
- Offline diagnosis requests need to be handled when connectivity returns.
- User-generated content requires moderation.
- Security and activity logging must not unnecessarily expose sensitive user information.
- Requirements should remain technically achievable within the project scope.

5. Farmer Representative

What does this stakeholder want from the system?

- A simple application that can help farmers with everyday agricultural problems.
- Easy access to crop health information.
- Quick assistance when a crop appears unhealthy.
- Practical treatment and prevention advice.
- Weather information relevant to their location.
- The ability to ask agricultural questions.
- Access to other farmers' experiences and knowledge.
- The ability to use important features even when internet connectivity is poor.

What features do they consider important?

- Bengali language support.
- Simple registration and login.
- Automatic location detection.
- AI-based crop diagnosis.
- Crop disease treatment and prevention advice.
- Diagnosis history.
- Location-based weather forecast.

- Agricultural weather alerts.
- Crop information library.
- Offline crop information.
- Offline diagnosis queue.
- Farmer community forum.
- Image sharing.
- Comments and likes.
- Notifications.
- AI agricultural chatbot.
- Fertilizer and farming-related calculations/recommendations.

What constraints or concerns do they have?

- Some farmers may have limited technical knowledge.
- Internet connectivity may be unreliable in rural areas.
- The interface should be simple and easy to understand.
- Bengali language support is important for accessibility.
- AI recommendations should not be misleading.
- Farmers may need practical answers rather than complicated technical explanations.
- Location information should only be accessed with user permission.
- Personal information and uploaded crop images should be protected.

6. Agricultural Expert

What does this stakeholder want from the system?

- A reliable platform for providing agricultural guidance.
- Access to relevant farmer questions and crop information.
- An opportunity to review or verify agricultural recommendations.
- A way to provide professional guidance when AI-generated assistance is insufficient.
- A mechanism for maintaining the quality of agricultural information.

What features do they consider important?

- Expert access.
- Crop diagnosis information.
- Farmer questions.
- Community forum.
- AI chatbot information that may require expert verification.

- Crop information library.
- Treatment and prevention recommendations.
- Regulatory oversight.
- Quality assurance.
- Input verification.
- Notifications.
- Activity records where appropriate.

What constraints or concerns do they have?

- AI-generated diagnosis may contain uncertainty.
- AI recommendations should not automatically be treated as professional agricultural advice.
- Agricultural information should be verified before being presented as authoritative.
- Recommendations should be appropriate for the crop and farming context.
- Expert involvement should be possible when automated assistance is insufficient.
- Farmer information should remain confidential.

7. Weather Data Provider

What does this stakeholder want from the system?

- Weather information to be presented accurately and appropriately.
- Timely delivery of relevant weather information to farmers.
- Weather information to support agricultural planning.
- Proper representation of weather forecasts and warnings.

What features do they consider important?

- Automatic location detection.
- Location-based weather forecasts.
- Current weather information.
- Agricultural weather alerts.
- Heavy rainfall warnings.
- Other relevant weather notifications.
- Notification delivery.

What constraints or concerns do they have?

- Weather forecasts can change over time.

- Forecast information contains inherent uncertainty.
- Weather data must be presented with appropriate timestamps or update information.
- External data availability may affect the service.
- The system should avoid presenting outdated information as current.

8. Government Agriculture Officer

What does this stakeholder want from the system?

- A reliable agricultural support platform.
- Accurate agricultural information reaching farmers.
- A system that complements existing agricultural services.
- Appropriate quality control over agricultural information.
- Regulatory oversight of information provided through the platform.
- A mechanism for verifying important agricultural inputs and recommendations.

What features do they consider important?

- Crop information library.
- AI crop health diagnosis.
- Treatment and prevention advice.
- AI agricultural chatbot.
- Fertilizer and farming recommendations.
- Agricultural weather alerts.
- Expert consultation.
- Input verification.
- Quality assurance.
- Regulatory oversight.
- Content moderation.
- Security and activity logs.
- Bengali language support.

What constraints or concerns do they have?

- Agricultural recommendations should not contradict appropriate official guidance.
- Important agricultural information should be verified before being treated as authoritative.
- AI-generated information should be clearly distinguishable from verified government or expert guidance.

- Fertilizer and treatment recommendations should be provided responsibly.
- Farmer privacy and personal information must be protected.
- Community content should be monitored for harmful or misleading information.
- The system should complement existing agricultural support services rather than falsely represent itself as an official government service.

2.3 Working towards Collaboration

Common Requirements

Things nearly every stakeholder independently asked for:

- User registration/login with Bengali language support
- AI-based crop health diagnosis with treatment/prevention advice
- Diagnosis history
- Searchable crop information library
- Location-based weather forecasts and agricultural alerts
- AI agricultural chatbot
- Farmer community forum
- Notifications
- Content moderation
- Input verification / quality assurance for agricultural information
- Security and activity logging
- Clear distinction between AI-generated and expert verified information

Conflicting Requirements

- **Speed/availability of AI advice** (Farmer, Sponsor) **vs. mandatory expert verification before trust** (Expert, Govt Officer, HR Consultant). Verification adds delay that undermines "quick assistance" farmers want.
- **Full featured, comprehensive system** (Sponsor, HR Consultant, Govt Officer) **vs. limited time/resources and manageable scope** (PM, Dev Team). Feature wish-list exceeds academic project bandwidth.
- **Open, freely shared community forum** (Farmer) **vs. strict content moderation and regulatory oversight** (Govt Officer, Sponsor) .
- **Simple, low tech interface for low literacy users** (Farmer) **vs. robust security** (PM, Dev Team, Sponsor).
- **Automatic location detection for convenience** (Sponsor, Weather Provider) **vs. user privacy** (Farmer, Govt Officer).
- **Presenting weather data with full uncertainty** (Weather Provider) **vs. simple, non-technical answers** (Farmer) .
- **Offline first operation** (Farmer, Dev Team) **vs. real-time sync integrity and up to date verified content** (Sponsor, Expert) .

Final Requirements

- Bengali first UI with English toggle; simple, low literacy friendly design; optional SMS/voice fallback for feature-phone users
- AI crop diagnosis with a clearly labeled confidence/uncertainty indicator, routed to expert review when confidence is low
- Two-tier information system: "AI suggested" vs. "Expert verified" content, visually distinguished
- Offline mode limited to previously cached content + a diagnosis request queue that syncs and re-validates once online
- Location detection as opt-in only
- Weather forecasts displayed with timestamp
- Community forum with lightweight moderation

- Role based access (Farmer, Expert, Moderator, Admin) with security logging scoped to protect farmer data privacy
- Registration/login, AI diagnosis + history, weather alerts, crop library, chatbot, basic forum, admin dashboard and offline queue as core features, since these directly address the Expert/Govt accuracy concern
- Deferred items : market price info, mobile payment integration, supplier directory, dialect support noted as valuable but outside academic timeline.

Elicitation

3.1.1 Normal Requirements

Explicitly stated by stakeholders during elicitation which satisfy customers when present, dissatisfy when absent.

- User registration and login
- Bengali and English language support
- Automatic location detection (with permission)
- AI-based crop health diagnosis
- Crop disease treatment and prevention advice
- Diagnosis history
- Location-based weather forecasts
- Agricultural weather alerts
- Searchable crop information library
- Offline crop information and diagnosis queue
- Farmer community forum with image sharing, comments, and likes
- Notifications
- AI chatbot for agricultural questions
- Admin dashboard and user management
- Content moderation
- Security and activity logs

- Regulatory oversight, input verification, and quality assurance
- Role-based access for farmer, expert, moderator, and administrator

3.1.2 Expected Requirements

Fundamental like: rarely stated outright because stakeholders assume them, but their absence causes real dissatisfaction

- Data security and encryption for personal information and uploaded crop images
- System availability/uptime and graceful handling of external service failures (AI service, weather API downtime)
- Reasonable response time for diagnosis and chatbot queries
- Basic input validation and error handling (e.g., invalid image uploads, empty forms)
- Password recovery
- Consistent, timestamped data (so users know how current the weather/diagnosis info is)
- Cross-device/screen-size responsiveness for the app UI
- Data backup and recovery for user records and diagnosis history
- Clear permission prompts before accessing location, camera, or storage
- Sync conflict handling when offline queue reconnects
- Basic accessibility (readable font sizes, sufficient contrast) for older or low-literacy farmers
- Audit trail for admin/moderator actions (who approved/rejected what content)
- Session management
- Scalability to handle a growing number of users and records without major redesign

3.1.3 Exciting Features

- Personalized crop calendar with harvest reminders based on location and season
- Gamification of community participation (badges, reputation score for helpful farmers/experts)

4. Scenario-Based Modeling

4.1 Usage Scenarios

Scenario 1: User Registration and Login

Actor: Farmer

Goal: Access the Digital Farming Assistant securely.

The farmer opens the application and creates an account by providing the required information. After registration, the farmer logs in using their credentials and accesses the application's features.

Scenario 2: Crop Health Diagnosis

Actor: Farmer

Goal: Identify a possible crop disease or health problem.

The farmer takes or uploads a photo of an affected crop. The system analyzes the image and provides a possible diagnosis along with treatment and prevention advice. The result is saved in the farmer's diagnosis history.

Scenario 3: Weather Forecast and Alerts

Actor: Farmer

Goal: Obtain weather information relevant to farming.

The application detects the farmer's location with permission and provides the relevant weather forecast. If important weather conditions such as heavy rainfall are detected, the system sends an agricultural weather alert.

Scenario 4: AI Agricultural Chatbox

Actor: Farmer

Goal: Get quick answers to agricultural questions.

The farmer enters a question, such as the required fertilizer quantity for a specific field area. The AI chatbox processes the question and provides relevant agricultural guidance.

Scenario 5: Crop Information Search

Actor: Farmer

Goal: Learn about a specific crop.

The farmer searches for a crop in the information library. The system displays relevant information about cultivation, fertilizer, pests, diseases, and farming practices.

Scenario 6: Farmer Community

Actor: Farmer

Goal: Share information and discuss farming problems.

The farmer creates a community post and may attach a crop image. Other farmers can view the post, comment, and like it. The farmer receives notifications about relevant interactions.

Scenario 7: Expert Consultation

Actor: Farmer / Agricultural Expert

Goal: Receive professional agricultural guidance.

The farmer submits a question or crop-related problem to an agricultural expert. The expert reviews the information and provides appropriate recommendations through the platform.

Scenario 8: Offline Diagnosis Queue

Actor: Farmer

Goal: Use important features when there is no internet connection.

The farmer submits a crop diagnosis request while offline. The application stores the request and processes it when an internet connection becomes available.

Scenario 9: Administration and Content Moderation

Actor: Administrator

Goal: Manage and maintain the platform.

The administrator logs into the dashboard to manage users, agricultural content, experts, notifications, and reported community posts. The administrator can review and moderate inappropriate content.

Scenario 10: Agricultural Information Verification

Actor: Agricultural Expert / Government Agriculture Officer

Goal: Maintain the quality and reliability of agricultural information.

The authorized stakeholder reviews agricultural information, recommendations, or important inputs. The information can be verified, updated, or flagged when necessary before being treated as verified guidance in the system.

